American Computer Science League

2024-2025 ● Contest 3: Short Problem Solutions ● Elementary Division

1. Boolean Algebra

The Boolean expression $NOT\ (x \leq 24)\ OR\ (x\ <\ 23)$ is TRUE for the integers that are either greater than 24 or less than 23. They are 20, 21, 22, 25, 26, 27 which is 6 of the 8 numbers between 20 and 27 inclusive.

D. 6

2. Boolean Algebra

A Truth Table could be used, but it's easier to recognize that $\sim B\ *\ (A\ +\ B) = \sim B\ *\ A\ +\ \sim B\ *\ B = A\ *\ \sim B\ +\ 0 = A\ *\ \sim B$ which is only TRUE for the ordered pair (1,0).

C. (1,0)

3. Boolean Algebra

Filling in the Truth Table is as follows:

A	B	$A\ *\ B$	$\sim A$	$\sim A\ +\ A\ *\ B$	$\sim B$	$(\sim A\ +\ A\ *\ B)\ *\ \sim B$
0	0	0	1	1	1	1
0	1	0	1	1	0	0
1	0	0	0	0	1	0
1	1	1	0	1	0	0

The last column from top to bottom is 1000.

A. 1000

4. Boolean Algebra

Simplify the Boolean expression using the Distributive Property as follows:

$\sim A\ *\ (A\ *\ B\ +\ \sim A\ *\ \sim B)\ +\ \sim B\ *\ (A\ *\ \sim B\ +\ \sim A\ *\ B)$
$= \sim A\ *\ A\ *\ B\ +\ \sim A\ *\ \sim A\ *\ \sim B\ +\ \sim B\ *\ A\ *\ \sim B\ +\ \sim B\ *\ \sim A\ *\ B$
$= 0\ +\ \sim A\ *\ \sim B\ +\ \sim B\ *\ A\ +\ 0$
$= \sim B\ *\ (A\ +\ \sim A)$
$= \sim B\ *\ 1 = \sim B$

E. $\sim B$

5. Boolean Algebra If the Boolean algebra operator $A \oplus B$ is TRUE for the ordered pairs (0,1) and (1,0), then $\sim(A \oplus B)$ is TRUE for the ordered pairs (0,0) and (1,1). Each of the choices results in the following: a. $A * B + \sim A * \sim B$ is (1,1) or (0,0) which is correct. b. $A * \sim B + \sim A * B$ is (1,0) and (0,1) which is $A \oplus B$. c. $A + \sim B$ is (1,1), (1,0), (0,0). d. $B + \sim A$ is (1,1), (0,1), (0,0) e. $\sim A + \sim B + A + B$ is all 4 of them.	A. a
6. Boolean Algebra Teaching Math (M) and Science (S) or Social Studies (SS) and Language Arts (LA) is represented as $M * S + SS * LA$ given the words being used and the order of operations.	D. $M * S + SS * LA$